Comparative Study and Design of Current Starved Ring Oscillators in 16 nm Technology

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***Abstract*—In this brief, different characteristics of current starved ring oscillator (CSRO) circuits are investigated and four new designs are proposed. A new parameter, known as phase noise bandwidth product (PNBP), is also proposed to address the trade-off between phase noise performance and frequency tuning range while designing ring oscillators (ROs). The proposed circuits are compared with existing architectures in terms of power consumption, frequency bandwidth, maximum frequency, phase noise, power delay product (PDP) and PNBP. For performance evaluation, three-stage CSROs have been implemented in Cadence software using the 16 nm PTM High Performance (HP) technology model. Simulation results demonstrate that two newly proposed CSRO architectures using one NMOS sink or one PMOS source have similar optimum results in terms of PDP and PNBP as the existing CSRO using an output switch scheme, but with an advantage of lower area requirement. Moreover, all four proposed architectures show improved performance in terms of PDP and PNBP compared to conventional CSRO. The performance of the CSROs is also observed and compared with respect to the variations in process-voltage-temperature (PVT) parameters.**

***Index Terms*—CSRO, current source, current sink, PDP, PNBP, PVT variations.**

I. Introduction

THE VOLTAGE controlled oscillator (VCO) is an indispensable part of the phase lock loop (PLL) in the communication system for clock signal generation and frequency selection [1]. Some characteristics such as low power consumption, wide frequency ranges of operation and high integration capacity are important to be addressed when designing a VCO [2], [3]. A basic single ended ring oscillator (SERO) consists of a chain of an odd number of inverters with parasitic load capacitances at the output of every stage [4]. Each inverter stage introduces a delay on which the frequency of oscillation is dependent. The propagation delay can be controlled by the amount of current available to charge or discharge the capacitive load of each stage. The VCO's oscillation frequency is inversely proportional to the sum of the charging (t_{ch}) and discharging time (t_{dis}) [6] and is given by equation (1), where I_{bias} is the current through each stage, N is the number of stages, C_{tot} is the load capacitance of each stage and V_{DD} is the supply voltage.

$$f_{osc} = \frac{1}{N(t_{ch} + t_{dis})} = \frac{I_{bias}}{NC_{tot}V_{DD}} \tag{1}$$

Additional MOSFETs in series with the CMOS inverters can increase the equivalent resistance and limit the current flow. Such a circuit is known as a current starved ring oscillator (CSRO). For low power applications, the current starved (CS) delay stage is a common technique in SERO [5]. Although the CSRO provides lower power consumption, the reduced current causes a detrimental effect on the maximum frequency, tuning range, and linearity of the frequency-voltage characteristic. Different CSRO architectures have been presented in the literature [7]–[9] which use different biasing circuits. The delay stage can also be remodeled to minimize the demerits of the conventional CSRO. Another CSRO is presented in [10] which utilizes an output-switch scheme. A CSRO with only NMOS current sinks with no bias circuit is presented in [11] to reduce circuit complexity and power requirement. Jovanovic and Stojcev [12] presented a delay element with higher linearity than the conventional design.

The objective of the present research is to study the different CSRO architectures in a single technology environment and present new models of CSRO circuits. A new performance parameter, phase noise bandwidth product (PNBP), is also proposed to address the trade-off between phase and bandwidth while comparing different oscillator circuits. All the circuits are then compared in terms of power consumption, frequency bandwidth, maximum frequency, phase noise, power delay product (PDP) and PNBP. Simulations are done for three-stage CSROs using the 16 nm PTM High Performance (HP) technology model in Cadence Virtuoso software.

This brief has been divided into five sections, including Introduction. Section II presents the existing circuits in literature. In Section III, new CSRO architectures are discussed. Section IV presents simulation results and comparison of the different oscillator circuits and finally, conclusive statements are discussed in Section V.

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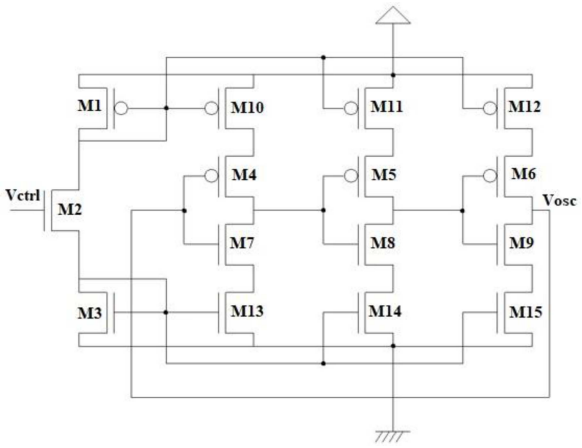


Fig. 1. Conventional CSRO.

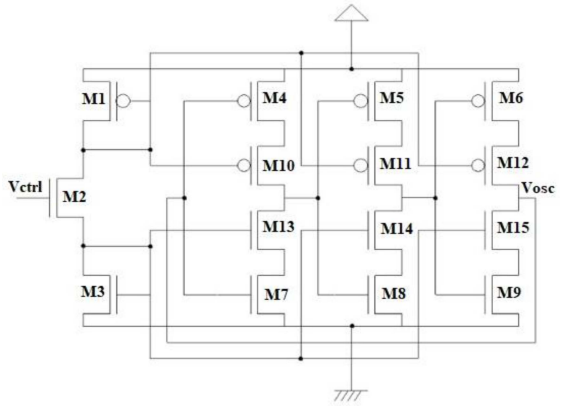


Fig. 2. CSRO with output switch scheme.

II. Existing CSRO Architectures

A. Conventional CSRO

The simplest conventional CSRO architecture is shown in Fig. 1 which includes a bias circuit (M1-M3), basic CMOS inverters (M4-M9), PMOS current sources (M10-M12), and NMOS current sinks (M13-M15). The bias current in (1), and hence the frequency, is controlled by the external control voltage V_{ctrl}. The current is generated from the bias circuit and is mirrored on to the inverters using the current mirror technique.

To keep the linearity for a wide range of frequency, the bias current must vary linearly with the change in control voltage. But in a conventional design for a particular range of control voltage, one of the transistors in the bias circuit (PMOS) operates in the saturation region while another (NMOS) operates in the linear region which creates non-linearity in the biasing current. The operating region of the current source/sink transistors (CSTs) also determines the linearity of the VCO. When the control voltage is high, the CSTs operate in the triode region and their gate voltages have less effect on the bias current [13]. The consequence is a reduction in the frequency tuning range. Moreover, the convergence of the drain voltages of M13-M15 with the output voltage of each stage prior to the discharging of load capacitance increases the delay.

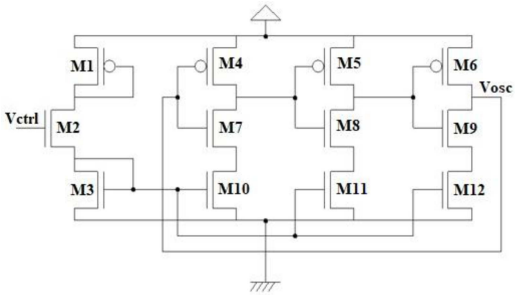


Fig. 3. CSRO with NMOS current sinks.

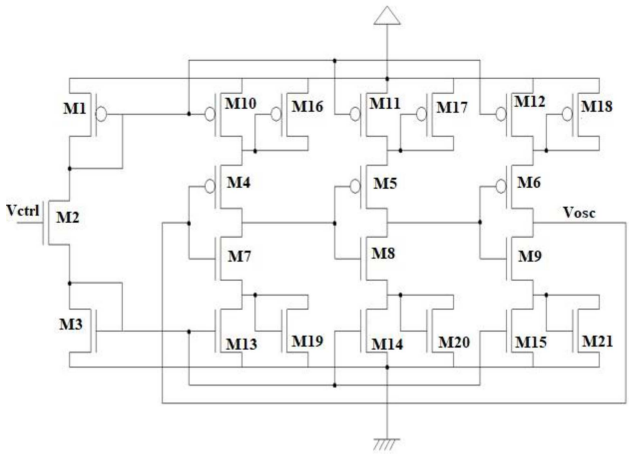


Fig. 4. RO with CS symmetric load transistors.

B. CSRO With Output Switch Scheme

In the CSRO circuit with the output-switch technique [10], the placement of the two control transistors in series with the CMOS inverter is different than the conventional CSRO, as shown in Fig. 2. Transistors M4-M9 make up the CMOS inverters and M10-M15 are the CS elements. During the rising input, the output capacitance always discharges from V_{DD}, and faster for the same current of control transistors, compared to the conventional circuit [8]. The discharging time is independent of the mismatch of the inverter transistors. The design technique showed very little impact of process variation compared to the conventional circuit.

C. CSRO With Only NMOS Sinks

To reduce the complexity, die area and power consumption, a CSRO is designed by Dinesh and Bharadwaj [11] using only NMOS sinks and eliminating PMOS sources and mirror elements. The current in each stage is only controlled by the NMOS sinks. The circuit provided good phase noise performance and linearity. The frequency was controlled by varying the V_{DD} instead of any separate control voltage. For the sake of uniform comparison in this brief, the bias circuit is included along with the NMOS sinks (M10-M12), as shown in Fig. 3. By the elimination of PMOS sources which reduces the overall impedance of each circuit, the current can be increased in order to increase the oscillating frequency.

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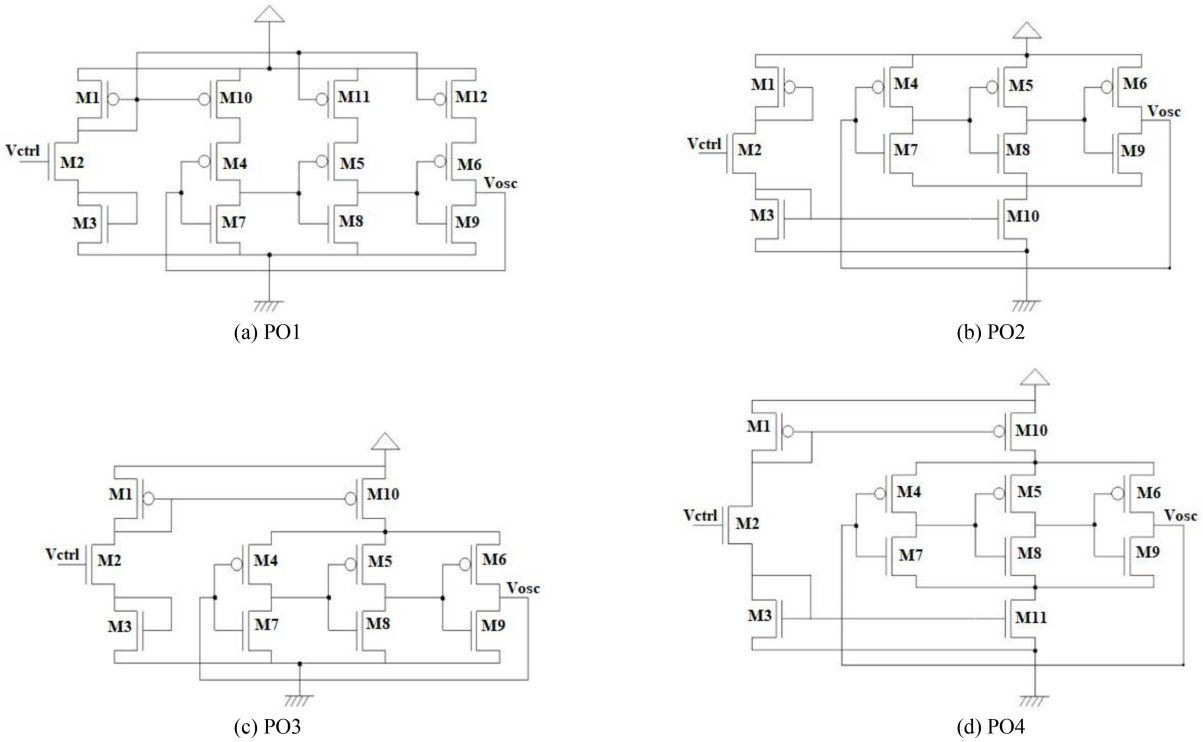


Fig. 5. Proposed 4 different CSRO architectures.

TABLE I
MOSFET DIMENSIONS FOR DIFFERENT CSRO ARCHITECTURES

Circuit	MOSFET widths (nm)		
	CMOS inverter	CS transistors	Bias circuit
Conventional CSRO	$W_n = 320$ $W_p = 640$	$W_n = 32$ $W_p = 64$	$W_n = 32$ $W_p = 64$
[12]		$W_n = 32$ $W_p = 64$	
[11]		$W_n = 96$ $W_p = 192$	
PO1		$W_n = 96$ $W_p = 192$	
PO2		$W_n = 96$ $W_p = 192$	
PO3		$W_n = 96$ $W_p = 192$	
PO4		$W_n = 96$ $W_p = 192$	
[10]	$W_n = 32$ $W_p = 64$	$W_n = 320$ $W_p = 640$	

D. RO With CS Symmetric Loads

Conventional CS delay elements are not suitable for a wide range of voltage regulation. Output propagation delay variation is not linear with respect to control voltage and that deteriorates the performance of the variable delay line. Therefore, a modified design is proposed [12] by adding transistors in parallel with CSTs as symmetric loads which helps to operate the transistors M10-M21 in saturation mode. As a result, the current through MOSFETs is now independent of the supply voltage which improves the stability with respect to supply noise and hence the phase noise performance. The involvement of symmetric load improved linearity at the cost of a lower range. The modification also results in increased charging and discharging current which increases the frequency of oscillation compared to the conventional design. However, for greater dimensions of source transistors than that of sink transistors, a larger amount of delay regulation is obtained. A three-stage RO with CS symmetric loads is shown in Fig. 4.

III. PROPOSED CSRO ARCHITECTURES

Four new CSRO circuits are presented and analyzed in this brief: (a) CSRO with only PMOS sources (PO1), (b) CSRO with 1 NMOS sink (PO2), (c) CSRO with 1 PMOS source (PO3) and (d) CSRO with one source-sink pair (PO4). The first circuit is developed as an alternate arrangement from the circuit in [11] where the NMOS current sinks are eliminated from the conventional circuit and transistors M10-M12 are the PMOS current sources, as shown in Fig. 5(a). For the second circuit configuration, the three individual NMOS current sinks of each stage are converted to one single sink for all three stages with a triple width dimension. The circuit is shown in Fig. 5(b) with M10 as the single NMOS sinking transistor. The third circuit is the alternate version of the second configuration, with one PMOS current source instead of the single current sink transistor. Figure 5(c) shows the CSRO topology with one PMOS current source transistor M10. Using one NMOS or PMOS reduces the required number of transistors and also the parasitic capacitance. The decreased capacitance is expected to improve the delay, and hence, the frequency performance of the circuit. The final proposed circuit consists of one NMOS current sink and one PMOS current source for the three stages combined. Transistors M10 and M11 in Fig. 5(d) make up the single CS pair for all three stages. The extra transistor in PO4 compared to PO2 and PO3 will improve the power requirement of the circuit at the cost of curtailed frequency.

IV. RESULTS AND ANALYSIS

A. Simulation Environment

All four existing circuits and four new circuits are simulated in 16 nm PTM-HP technology using the Cadence virtuoso

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TABLE II
PERFORMANCE OF DIFFERENT CSRO ARCHITECTURES

Architecture	Power (nW)	Maximum Frequency (MHz)	PDP (fJ)	Tuning Range (MHz)	Phase Noise @ 1MHz (dBc/Hz)	PNBP (kdB)
Conventional	167.30	38.68	4.32	34.29	78.71	2.70
[10]	167.50	340.24	0.49	255.67	73.23	18.72
[11]	230.20	157.06	1.47	128.98	71.39	9.21
[12]	169.06	43.48	3.89	27.95	78.92	2.21
PO1	232.40	173.4	1.34	136.86	74.69	10.22
PO2	261.20	332.64	0.79	276.50	66.41	18.36
PO3	244.90	329.22	0.74	262.29	69.24	18.16
PO4	181.80	170.78	1.06	146.10	75.76	11.07

platform. The biasing circuit configuration is fixed for all eight circuits for a fair comparison. The size of MOSFETs for the different sections in the circuits are given in Table I. The optimal sizing of transistors for the different architectures is not analyzed in this brief and, therefore, are not equally optimal. The same sizing is used for uniform comparison which provides oscillation in the chosen simulation environments. For the circuit in [10], the transistor sizing for inverter and CS elements are switched in order to obtain oscillation. This also ensures the same area requirement as the conventional circuit. The supply voltage V_{DD} is kept constant at 0.5 V while the control voltage V_{ctrl} is varied between 0.3 V and 1.0 V for frequency tuning.

B. Power and Frequency Performance

The performance of the circuits in terms of different features is presented in Table II. The power and maximum frequency parameters are calculated for the maximum control voltage of 1 V. Since the proposed circuits PO1-PO3 fewer number of transistors compared to the conventional circuit, the overall equivalent impedance is reduced which increases the current. According to (1), an increased current will raise the frequency (or decrease the delay) of the circuit. Moreover, in circuits PO2-PO4, the CS transistors of each stage are combined to a single transistor with its size tripled. The current through a MOSFET, given by (2), depends on the transistor size (W and L), carrier mobility μ, oxide capacitance C_{OX}, drain-source voltage V_{DS} (influenced by V_{DD}), gate-source voltage V_{GS} and the threshold voltage V_T. The transistor width also affects the magnitude of current indirectly through the threshold voltage, which is inversely dependent on the transistor size [14]. On the other hand, the capacitance of the single transistors in PO2-PO4 is now divided and shared among the 3 stages, which eventually reduces the total parasitic capacitance. Therefore, the oscillating frequency of the proposed circuits is significantly increased due to these phenomena compared to the conventional circuit.

$$I_{bias} = \mu C_{OX} \frac{W}{L}\left[(V_{GS} - V_T)V_{DS} - \frac{{V_{DS}}^2}{2}\right] \tag{2}$$

Simultaneously, an increased current means a rise in power consumption of the circuit, and hence, a trade-off exists between frequency and power while designing an RO. This relation is also visible from the results in Table II. Therefore, it is important to compare the performance of different oscillators in terms of the PDP to address this trade-off and indicate which architecture provides the optimum result. The PDP is a product of the power consumption and the propagation delay of the circuit. The propagation delay of the circuit is the inverse of the oscillating frequency, so the PDP is calculated using (3) at $V_{ctrl} = 1$ V and $V_{DD} = 0.5$ V. The units are mentioned in brackets. Table II presents the PDP performance for the different architectures. A lower PDP value is desired for a fast and power-efficient circuit.

$$PDP(\text{fJ}) = \frac{power(\text{nW})}{maximum\ frequency\ \ (\text{MHz})}. \tag{3}$$

C. Bandwidth and Phase-Noise Performance

The frequency tuning range and the phase noise performance of the different CSRO architectures are also presented in Table II. For the tuning range, the control voltage is varied between 0.3 V and 1 V. The phase noise is calculated at 1 MHz offset frequency for $V_{ctrl} = 1$ V. The absolute values of the phase noise (actual value is negative) are given; a higher value indicates better noise performance.

A comparison between the tuning range and phase noise values in the table shows the presence of another important trade-off in ROs, i.e., the noise performance is adversely affected by the tuning range of an oscillator. As the frequency tuning range increases, the absolute value of phase noise decreases. In order to address this trade-off, a new parameter is proposed which is a product of the absolute phase noise and the frequency bandwidth of the circuit and is calculated using (4). A high value of PNBP is desired for a circuit with a wide tuning range and good phase noise performance. Therefore, circuits with higher PNBP values provide better optimization of the trade-off. This parameter is, therefore, equally important as the PDP while designing or selecting the appropriate RO circuit.

$$PNBP(\text{kdB}) = \frac{phase\ noise\ \left(\frac{\text{dB}}{\text{Hz}}\right)\ \text{x}\ tuning\ range\ (\text{MHz})}{1000}. \tag{4}$$

D. Response to PVT Variation

The current through a transistor is influenced by V_{DD}, V_T and C_{OX}. The oxide capacitance C_{OX} is inversely proportional to the oxide thickness T_{OX}. Meanwhile, V_T and μ are temperature dependent. Therefore, it is important to observe the variation of the oscillating frequency in response to the process

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TABLE III
EFFECT OF PVT VARIATIONS ON OSCILLATING FREQUENCY

Parameter		T	V_{DD}	V_{TN}	V_{TP}	T_{OXN}	T_{OXP}
Ideal values		27°C	0.5V	0.48V	-0.43V	0.95nm	1nm
Frequency Variation (%)	Conventional	0.80	125.3	123.8	104.1	59.07	51.37
	[10]	-1.47	112.5	93.1	85.8	46.59	45.05
	[11]	1.27	121.6	129.9	31.8	60.02	53.63
	[12]	1.33	119.8	104.6	110.2	60.03	53.31
	PO1	1.21	123.4	105.6	121.8	44.98	61.15
	PO2	0.96	123.69	139.5	64.6	64.37	42.13
	PO3	0.94	125.21	98.2	128.5	43.13	63.40
	PO4	1.76	114.43	126.8	109.1	60.50	55.23

parameters, temperature and supply. The process parameters identified are threshold voltage of NMOS transistors V_{TN}, threshold voltage of PMOS transistors V_{TP}, oxide thickness of NMOS transistors T_{OXN}, and oxide thickness of PMOS transistors T_{OXP} [15].

The process parameters, supply voltage and temperature are varied between -10% to $+10\%$ of their values. The frequency response with respect to the change is presented in Table III. The frequency variation is calculated using (5), where the tuning range is the total variation of frequency and the center frequency is evaluated using the ideal values.

$$\%Frequency\ variation = \frac{tuning\ range}{center\ frequency} \tag{5}$$

Simulation results in Table III show that each circuit has improved sensitivity to a different parameter than the other circuits. The proposed circuits PO2 and PO3 have better temperature stability than the circuits in [10]–[12]. The circuit in [10] shows the highest stability with respect to the supply and V_{TN}, while the circuit in [11] shows the highest stability with respect to V_{TP}. The proposed circuits PO2 and PO3 show the least variation in the frequency with respect to T_{OXP} and T_{OXN}, respectively.

V. CONCLUSION

Different CS delay stage topologies exist in literature and this brief investigates the possibility of four new CSRO architectures. In total, eight different circuits are compared in terms of key features such as maximum frequency, tuning range, phase noise, and power consumption. In order to consider the trade-offs between these parameters, different oscillators can be compared with respect to two important features, the well-known PDP and the newly proposed PNBP. Simulation results show that the existing CSRO architecture with the output switch technique provides the best performance in terms of PDP and PNBP. However, proposed CSRO architectures PO3 and PO4 provide almost equal performance in terms of PDP and PNBP with a much lower count of transistors (5 fewer transistors) and hence, lower area requirement. These two new circuit techniques provide very good performance in terms of frequency, though they have the highest and lowest power requirement and phase noise value, respectively. The proposed circuit PO1 provides lower PDP value and higher PNBP value compared to the existing CSRO technique with only NMOS sinks and proves to be a better alternative. Although RO with CS symmetric loads provides the best phase noise performance and considerably low power consumption, it suffers from very poor frequency and bandwidth performance. All four proposed architectures give much better results in terms of PDP and PNBP compared to the conventional CSRO circuit. With respect to the variations in PVT parameters, proposed circuit PO2 shows good stability in terms of temperature, V_{TP} and T_{OXP}, while PO3 shows good stability in terms of temperature, V_{TN} and T_{OXN}.

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